Assignment 7 — Transfer Learning

Deep Learning | Practical Report / Solution

BRACT's Vishwakarma Institute of Technology, Pune

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Problem Statement

Implement Transfer Learning using pre-trained AlexNet, VGG16, ResNet50 and EfficientNetB0 models for image classification and compare their performance.

Objective

To demonstrate transfer learning with ImageNet-pretrained convolutional models, replace their final classification layers for a 10-class image classification task, evaluate test accuracy, record training time, and compare the models.

Algorithm / Methodology

1. Load the CIFAR-10 dataset and separate it into training and testing sets.
2. Resize and normalize images according to the input requirements of the pre-trained models.
3. Load the pre-trained AlexNet, VGG16, ResNet50, and EfficientNetB0 models.
4. Replace the final classification layer with a new layer having 10 output classes.
5. Freeze the pre-trained layers and train the newly added classification layers.
6. Evaluate each model on the test dataset and calculate classification accuracy.
7. Record the training time of each model.
8. Compare the models using a results table and accuracy comparison graph.

Model Architecture

The transfer-learning setup keeps the pretrained feature extractor frozen and adapts the classification head to the 10-class task. In the supplied notebook, VGG16, ResNet50 and EfficientNetB0 use TensorFlow/Keras ImageNet weights, while AlexNet uses torchvision because TensorFlow/Keras does not provide an official pretrained AlexNet implementation.

Notebook Implementation Details

The supplied Assignment_7.ipynb currently implements the experiment using a smaller MNIST subset rather than CIFAR-10: 10,000 training images and 2,000 test images are used after converting grayscale images to 3-channel RGB and resizing them to 32×32. VGG16, ResNet50 and EfficientNetB0 are trained for 3 epochs with batch size 64; AlexNet uses a frozen feature extractor and a 10-class classifier. This differs from the attached practical sheet, which specifies CIFAR-10.

Important source consistency note: The report preserves both supplied sources rather than silently changing one to match the other. The performance table below is the table explicitly provided in the practical sheet.

Performance Comparison

Model	Test Accuracy	Training Time (s)
AlexNet	73.6%	149.82
VGG16	74.4%	2790.55
ResNet50	69.9%	1150.35
EfficientNetB0	70.9%	427.66

According to the supplied practical-sheet results, VGG16 achieves the highest test accuracy at 74.4%. AlexNet has the lowest training time at 149.82 seconds among the four listed models.

Evaluation and Comparison

The primary comparison metrics are test accuracy and training time. Higher test accuracy indicates better classification performance, while lower training time indicates lower computational cost for the reported experiment.

Conclusion

Transfer learning was implemented using AlexNet, VGG16, ResNet50, and EfficientNetB0 for image classification. Based on the performance values supplied in the practical sheet, VGG16 achieved the highest test accuracy of 74.4%, followed by AlexNet at 73.6%, EfficientNetB0 at 70.9%, and ResNet50 at 69.9%. AlexNet also required less training time than the other models. Thus, the experiment demonstrates how different pre-trained models vary in classification accuracy and computational time.

Key Learning Outcomes

• Understand the idea of transfer learning and frozen pretrained feature extractors.
• Adapt pretrained networks to a new 10-class classification task.
• Compare architectures using accuracy and training time.
• Understand that architecture performance depends on the dataset, preprocessing, and training setup.

Source Files

Assignment_7.ipynb — supplied implementation notebook.
CSAIE_64_DL_Assignment 7.docx — supplied practical implementation sheet.